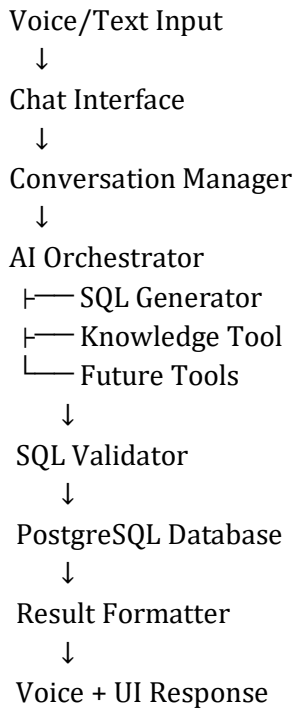


Enterprise Voice AI Agent - Team Plan (6 Members)

Goal: Build an enterprise voice AI assistant capable of understanding natural language, generating safe SQL queries, querying a cloud database, and responding with voice and visual insights by Friday evening.

Overall Architecture



Team Responsibilities

Member 1 – AI Orchestrator (Team Lead) (Me)

- Responsibilities:
 - Design overall architecture
 - Integrate all modules
 - Prompt engineering
 - Conversation memory
 - Intent detection
 - Tool routing
 - Final testing
- Deliverables:
 - orchestrator.py
 - tool_router.py

- memory.py
- prompt_builder.py

Member 2 – Voice AI

- Responsibilities:
 - Streaming STT
 - Streaming TTS
 - Voice Activity Detection
 - Microphone handling
 - Voice interruption
- Deliverables:
 - voice_service.py
 - stt.py
 - tts.py

Member 3 – Chat UI

- Responsibilities:
 - Enterprise chat UI
 - Streaming responses
 - Voice button
 - Animations
 - Charts
 - Conversation history
- Deliverables:
 - Flutter screens
 - Widgets
 - Animations

Member 4 – SQL Intelligence

- Responsibilities:
 - Schema loader
 - SQL generation
 - SQL validation
 - Safe query execution
 - Read-only database access
- Deliverables:
 - sql_engine.py

- validator.py
- executor.py
- schema_loader.py

Member 5 – Backend & APIs

- Responsibilities:
 - FastAPI
 - Authentication
 - API endpoints
 - WebSockets
 - Logging
 - Database connectivity
- Deliverables:
 - main.py
 - api.py
 - routes.py

Member 6 – Intelligence Layer

- Responsibilities:
 - Response formatting
 - Charts
 - Analytics
 - Follow-up suggestions
 - Conversation summaries
 - Testing & demo preparation
- Deliverables:
 - formatter.py
 - analytics.py
 - chart_generator.py

Development Timeline

Day	Focus
Monday	Finalize architecture, folder structure, API contracts, prompts, database schema.
Tuesday	Independent development of assigned modules.
Wednesday	Integration only. Connect all modules together.
Thursday	Performance tuning, bug fixes, charts, conversation memory, polish.

Friday	Demo preparation, rehearsals, edge-case testing, presentation.
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Recommended Repository Structure

```
enterprise-agent/  
├── frontend/  
├── backend/  
├── ai/  
│   ├── orchestrator.py  
│   ├── prompts.py  
│   ├── memory.py  
│   └── router.py  
├── sql/  
│   ├── validator.py  
│   ├── executor.py  
│   └── schema.py  
├── voice/  
│   ├── stt.py  
│   └── tts.py  
├── analytics/  
│   ├── charts.py  
│   └── formatter.py  
└── docs/
```

Key Advice

- Define interfaces between modules before coding.
- Use JSON contracts between components.
- Keep SQL execution read-only.
- Focus on a polished end-to-end demo rather than many incomplete features.
- Team lead should prioritize integration over implementing every feature.